Mock Final Examination

Semester I, Academic Year 2021-22

Student Name			
Student ID			
Department			
Section		**Venue** :	

Course Code & Name	ITDB101 Introduction to Database		
Specialisation	IT	**Level**	Diploma 1
Date	23 December 2021	**Time**	2 hrs

INSTRUCTIONS TO CANDIDATES

- Do not open this question paper until told to do so.
- Fill in the above details.
- Use black or blue pen only. Pencil may be used for drawing or graph.
- Electronic devices are not allowed in the examination venue.
- Scientific calculators can be used if only allowed by the course teacher.
- Copying or cheating or malpractices of any kind are strictly prohibited. Violations will be promptly reported to higher authorities, and actions will be taken.
- Verify the total number of answer sheets in this examination (8 pages including cover page).

FOR VALUATION PURPOSE ONLY

Section	Marks Allocated	Marks Scored	
A	15		
B	15		
C	20		_____
Total Marks	50		**50**

Marked by:

Name_____ Signature _____ Date _____

Re-evaluated by:

Name _____Signature _____ Date _____

SECTION A (15 * 1 = 15 Marks)

Choose the correct option.

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1. Raw, unorganized facts that need to be processed is called _____.

   a) Database Management System b) Data

   c) Information d) Database

2. The database user whose job requires access to database for querying, updating and generating reports is known as.

   a) Application Programmer b) Application user

   c) End user d) Sophisticated user

3. In advanced software applications like geographical information system, we use

   a) ER Diagrams b) EER Diagrams

   c) System Diagram d) Client Diagram

4. An entity named course has an attribute section, section has values section1, section2, section3, find the type of attribute.

   a) Simple Attribute b) Multivalued Attribute

   c) Stored Attribute d) Composite Attribute

5. Identify the relationship that involves only two entity types.

   a) Unary b) Binary

   c) Ternary d) Mandatory

6. Choose the property of ISA relationship that represents all attributes of super type entity apply to the subtype entity

   a) Entity b) Cardinality

   c) Inheritance d) Degree

7.A _____ is a special symbol independent of data type.

   a) Attribute b) Key c) Null d) Primary key

8._____ requires that every table has a primary key.

   a) Referential Integrity b) Entity Integrity
   c) Domain Integrity d) Data Integrity

9.What do you mean by one to many relationships?

   a. One class may have many teachers
   b. One teacher can have many classes
   c. Many classes may have many teachers

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d. Many teachers may have many classes

10. Which of the following is a top-down approach in which the entity's higher level can be divided into two lower sub-entities??

a. Aggregation
b.Generalization
c.Specialization
d.All of the above

11.Who created the first DBMS?

a) Edgar Frank Codd
b) Charles Bachman
c) Charles Babbage
d) Sharon B. Codd

12. The property / properties of a database is / are:

a)It is an integrated collection of logically related records.
b)It consolidates separate files into a common pool of data records.
c)Data stored in a database is independent of the application programs using it.
d)All of the above

13.E-R model uses this symbol to represent weak entity set?

a) Dotted rectangle.
b) Diamond.
c)Doubly outlined rectangle.
d)None of these.

14.Key to represent relationship between tables is called

a)Primary key
b)Secondary Key
c)Foreign Key
d)None of these

15.What is a relationship called when it is maintained between two entities?

a)Unary
b)Binary
c)Ternary
d)Quaternary

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SECTION B (5 * 3 = 15 Marks)

Short Answer Questions

11. What is data redundancy? Write any one of the problems created by data redundancy

12.Explain the following with suitable EER Diagram

(i) Specialization
(ii) Generalization

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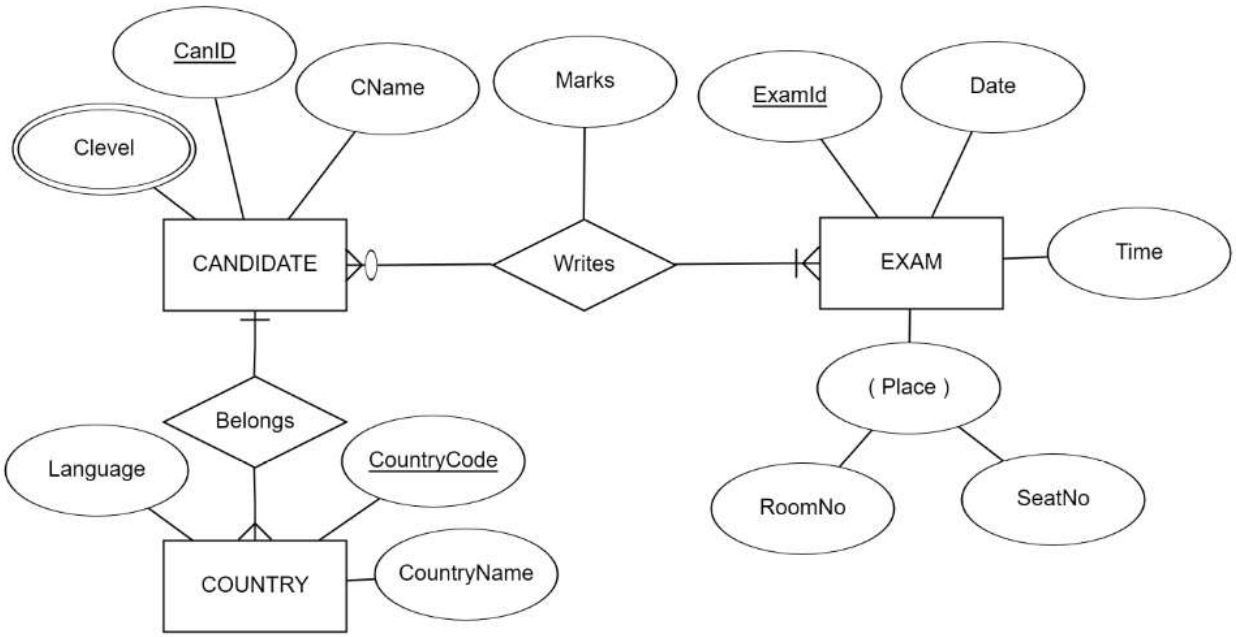
13.Differentiate between multivalued and derived attribute with suitable diagrams

14.Consider the following ER-diagram



Identify the following types of attributes and write one name of each types in the following table

Attribute type	Name
Derived	
Multivalued	
Week Entity	
Relationship	
Composite	
Identifier	

15.What is data redundancy? Write any one of the problems created by data redundancy

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SECTION C (2 * 10 = 20 Marks)

Long Answer Questions

16. (a) Normalize the below table into UNF, 1NF, 2NF and 3NF. Show primary keys and foreign keys (6 Mark)

StudID	StudName	SAddress	ProjID	ProjName	DeptID	DeptName	Super visor	Mark
S101	Hanifa	Mulladha	P10	Wills	IT	InfoTech	C1	65
			P11	Ruqaya	EN	Engg	C3	82
S202	Syed	Barka	P22	Ali	IT	InfoTech	C1	76
			P11	Ruqaya	EN	Engg	C3	69

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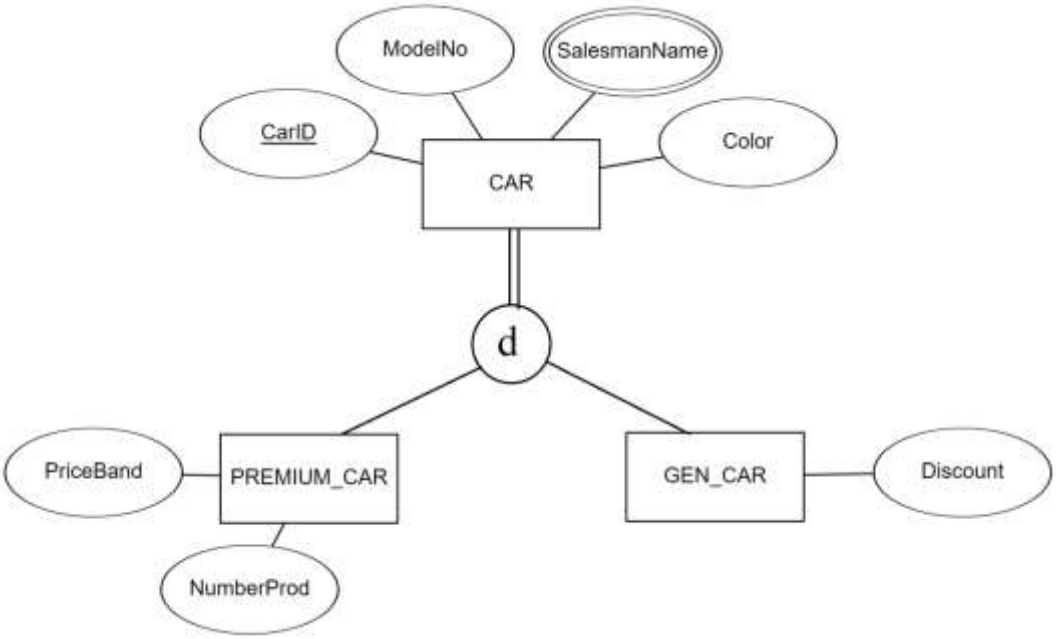
(b) Realize the below EER diagram and answer the following questions (4 Mark)



a. Write down name/s of subclasses shown in the diagram_____ (1)

b. Write down the name/s of a superclass shown in the department _____(1)

c. What type of participation is there of entities `PREMIUM_CAR` and `GEN_CAR` in the entity set `CAR`? _____. (1)

d. What type of specialization is shown for `PREMIUM_CAR` and `GEN_CAR` of the entity set `CAR`? _____. (1)

e. Write down the multivalued attribute shown in the diagram. _____

f. How many primary keys are there in the diagram? Give name. Number: _____

Name _____ (1)

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17. (a) Design a database to keep track of information (6 Mark)

In Oman Al Karim real estate company has many HOUSES which are given on rent to various TENENTS. A TENENT has attributes TenID as Identifier, BirthDate and Address, Address is composed of Street, City and State. TENENT can be union of TEACHER, STUDENT and TECHNITIAN. Attributes of TEACHER are TeachID(Identifier) and TName, attributes of STUDENT are StudID(Identifier) and SName, attributes of TECHNITIAN are TechId(Identifier) and TechName. TENENT can be separated as EXPATRIAT or NATIONAL who are disjoint and their participation is full and complete. Attributes of EXPATRIAT are SsID(Identifier) and ConName, only attribute for NATIONAL is NationalID.(Identifier).

Many HOUSES are rented to many TENENTS. HOUSE can be FLAT, VILLA and FARMHOUSE which are union but not disjoint entities. Attributes of HOUSE are HomeNo(Identifier), RentPrice and Model. Attributes of FLAT are FlatNo(Identifier) and Fstyle, attributes of VILLA are VillaNo(Identifier) and Vstyle, attributes of FARMHOUSE are FarmNo(Identifier) and Fstyle.

Draw an Enhanced Entity Relationship Diagram (EERD) for the above scenario, identify entities, attributes, subsets, union and disjoint entities, key and generalization and Specialization constraints if any.

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(b) Explain the following (4 Mark)

1. Functional Dependency
2. Difference between SQL and No SQL databases